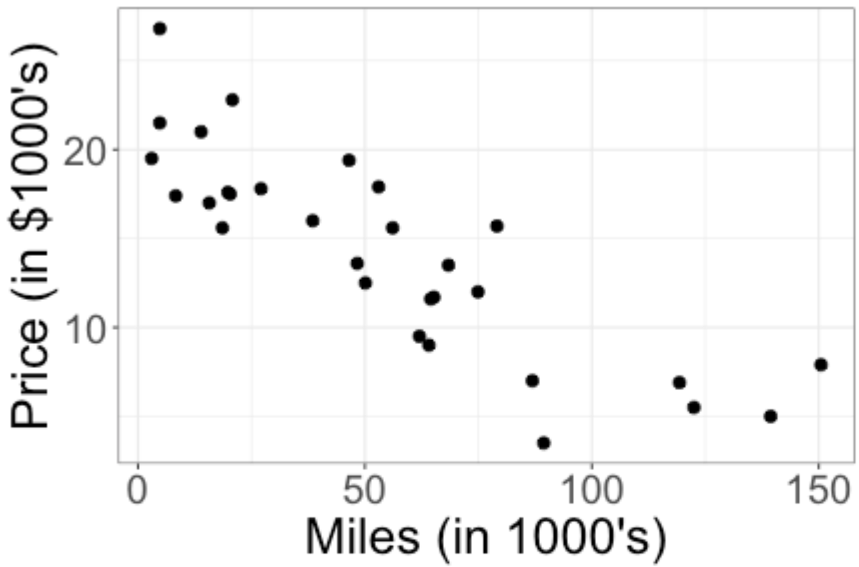
Inference for linear regression slope

Inference

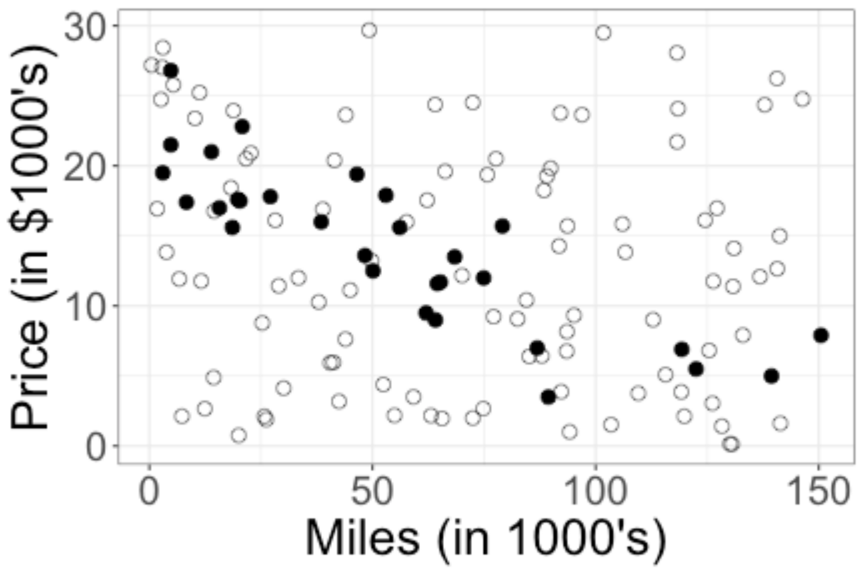
Is there a relationship between mileage and price for used Honda Accords?



It looks like it, but is that just a fluke?

Is the observed relationship a fluke?

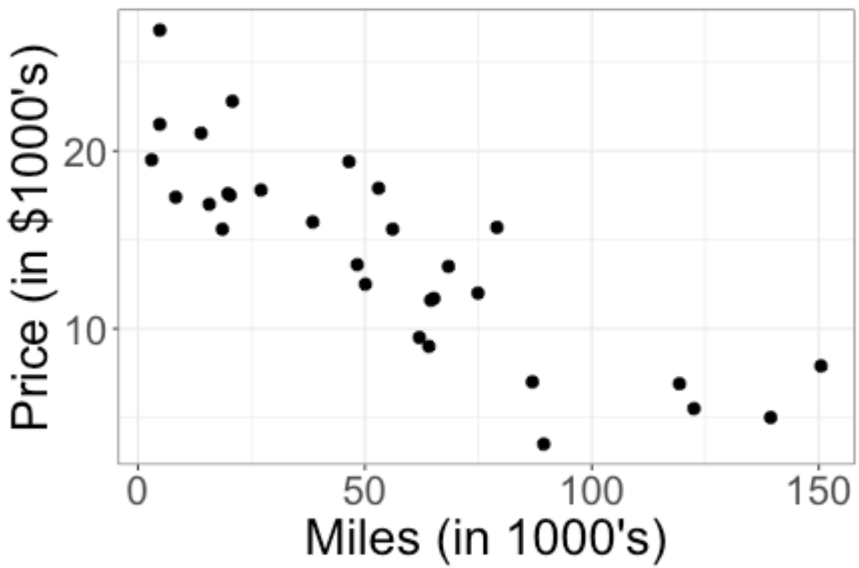
What if the population scatterplot looks like this?



Translating the question

if there is no relationship,
then $\beta_1 = 0$

Is there a relationship between mileage and price?

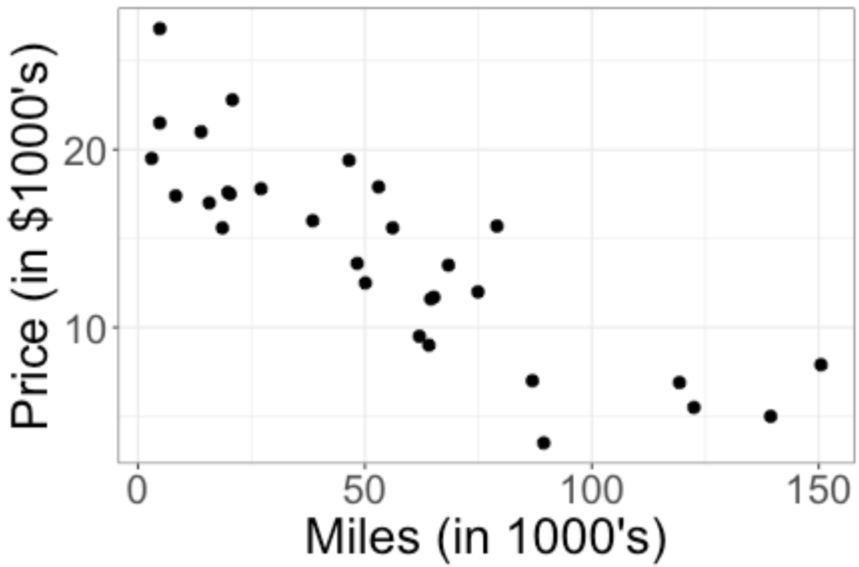


Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

> If there is no relationship between mileage and price, what is true about my model?

Translating the question

Is there a relationship between mileage and price for used Honda Accords?

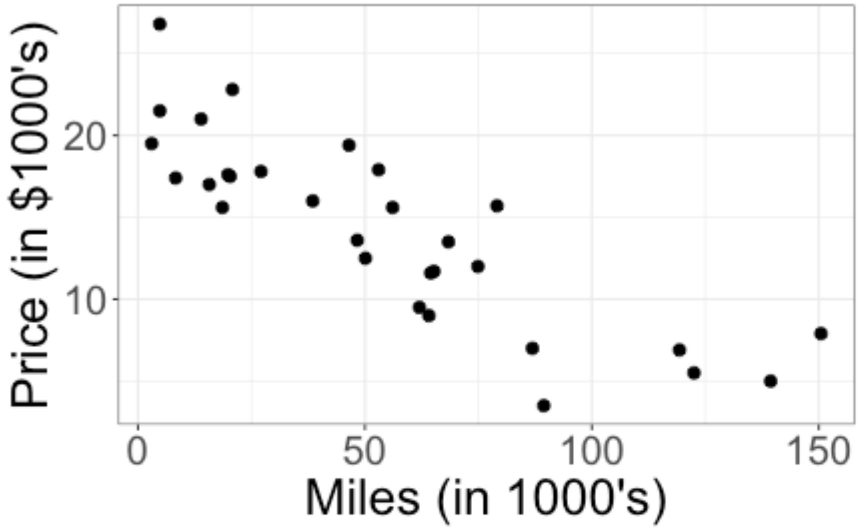


Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Question: Is $\beta_1 = 0$?

Hypothesis testing

$\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon \qquad \text{Is } \; \beta_1 = 0 \, ?$



In statistics, we try to address these kinds of questions with *hypothesis testing.*

Hypothesis testing

What steps does a hypothesis test involve?

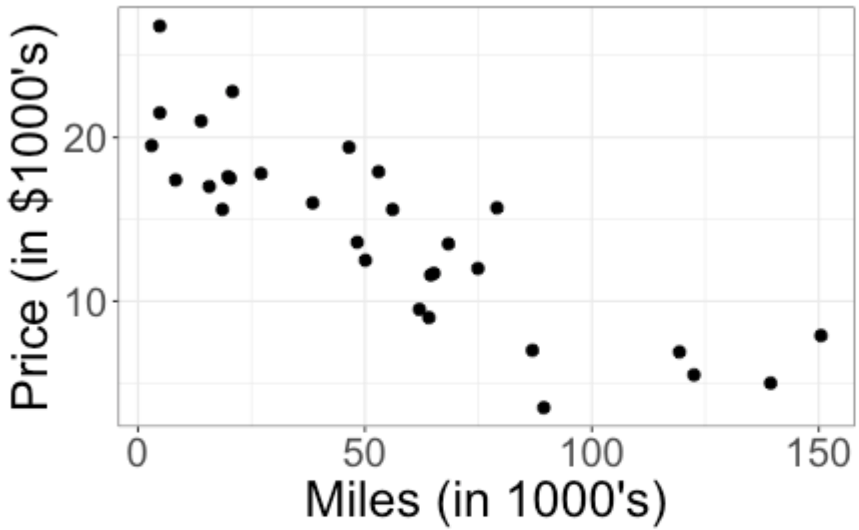
- specify hypotheses : null hypothesis (H_0)
  alternative hypothesis (H_A)
- check conditions
- If we are doing significance testing, then need to specify significance level (α)
- calculate test statistic
- p-value (need a distribution for test statistic under H_0)

H_0: $\beta_1 = 0$

H_A: $\beta_1 \neq 0$

Hypothesis testing

$\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon \qquad \text{Is } \beta_1 = 0\,?$



In statistics, we try to address these kinds of questions with *hypothesis testing*.

> What do you think our null hypothesis (H_0) and alternative hypothesis (H_A) should be?

Specifying hypotheses

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Question: Is $\beta_1 = 0$?

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

- The *null hypothesis*, H_0, is that there is no relationship
- The *alternative hypothesis*, H_A, is that there is a relationship
- Our hypotheses are stated in terms of one or more model parameters

Specifying hypotheses

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

> What if I want to know whether there is a *positive* relationship between mileage and price?

Specifying hypotheses

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

> What if I want to know whether there is a *positive* relationship between mileage and price?

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 > 0$$

Estimating the relationship

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Estimated line: $\widehat{\text{price}} = 20.8 - 0.12 \text{ miles}$

> If $\beta_1 = 0$, how unusual would $\widehat{\beta}_1 = -0.12$ be?

Estimating the relationship

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Estimated line: $\widehat{\text{price}} = 20.8 - 0.12 \text{ miles}$

> If $\beta_1 = 0$, how unusual would $\widehat{\beta}_1 = -0.12$ be?

> Need to know how $\widehat{\beta}_1$ varies from sample to sample. How do we measure the variability of a statistic?

How unusual is the observed slope?

$$H_0 : \beta_1 = 0$$

Standardize: $$t = \frac{\widehat{\beta}_1 - 0}{SE_{\widehat{\beta}_1}}$$

where $SE_{\widehat{\beta}_1}$ is the estimated standard error (standard deviation) of $\widehat{\beta}_1$

How unusual is the observed slope?

$$H_0 : \beta_1 = 0$$

Standardize: $$t = \frac{\widehat{\beta}_1 - 0}{SE_{\widehat{\beta}_1}}$$

Accord data:

```
accord_lm <- lm(Price ~ Mileage, data = AccordPrice)
summary(accord_lm)
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096     0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198     0.0141   -8.50 3.06e-09 ***
```

$\hat{\beta}_1$

$$t = \frac{-0.12 - 0}{0.0141} = -8.50$$

How unusual is the observed slope?

$$H_0 : \beta_1 = 0$$

Standardize: $$t = \frac{\widehat{\beta}_1 - 0}{SE_{\widehat{\beta}_1}}$$

Accord data:

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   20.8096     0.9529   21.84  < 2e-16 ***
## Mileage       -0.1198     0.0141   -8.50 3.06e-09 ***
```

> What does a value of $t = -8.50$ mean?

How unusual is the observed slope?

Accord data:

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096     0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198     0.0141   -8.50 3.06e-09 ***
```

The observed value of $\widehat{\beta}_1$ (-0.12) is 8.5 standard deviations below the value of β_1 hypothesized in $H_0 : \beta_1 = 0$

> Is 8.5 standard deviations a lot?

How unusual is the observed slope?

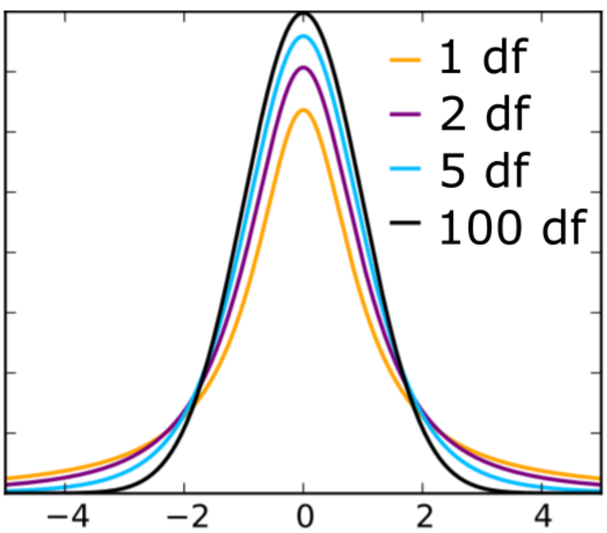
observations in data

parameters in model (β_0, β_1)

$$t = \frac{\widehat{\beta}_1 - 0}{SE_{\widehat{\beta}_1}}$$

If $\beta_1 = 0$, then t follows a t-distribution with $n - 2$ degrees of freedom (df)

t-distributions



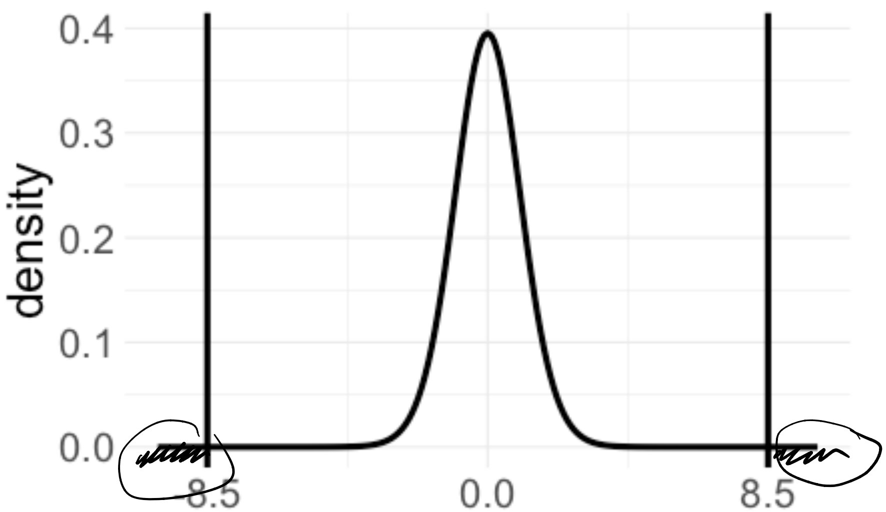
as df → ∞, t distribution approaches N(0,1)

for small df, t distribution has heavier tails (more variability)

How unusual is the observed slope?

$$t = \frac{-0.12}{0.014} = -8.5, \quad n - 2 = 28$$

t_{28} distribution:



$H_A: \beta_1 \neq 0$

$\Rightarrow$ two-tailed test

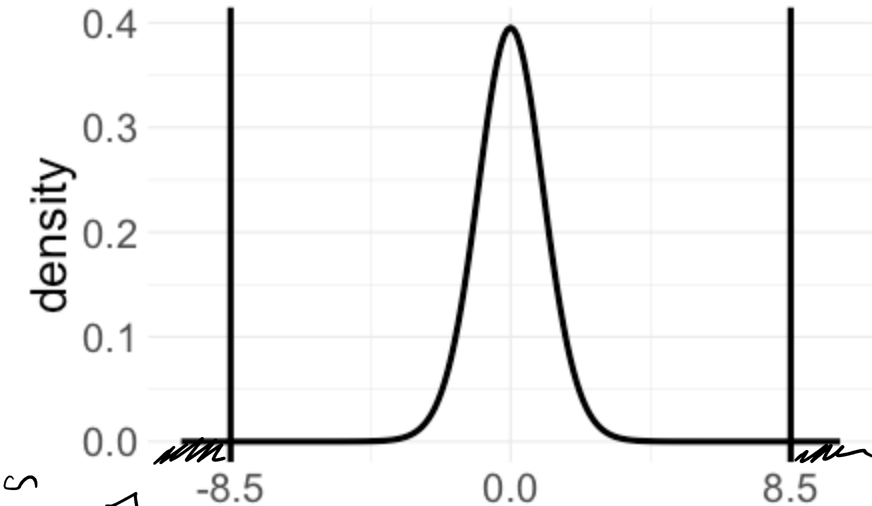
$H_A: \beta_1 > 0$

$\Rightarrow$ look in right tail

$\beta_1 < 0$

$\Rightarrow$ look in left tail

How unusual is the observed slope?



probabilities for t distribution

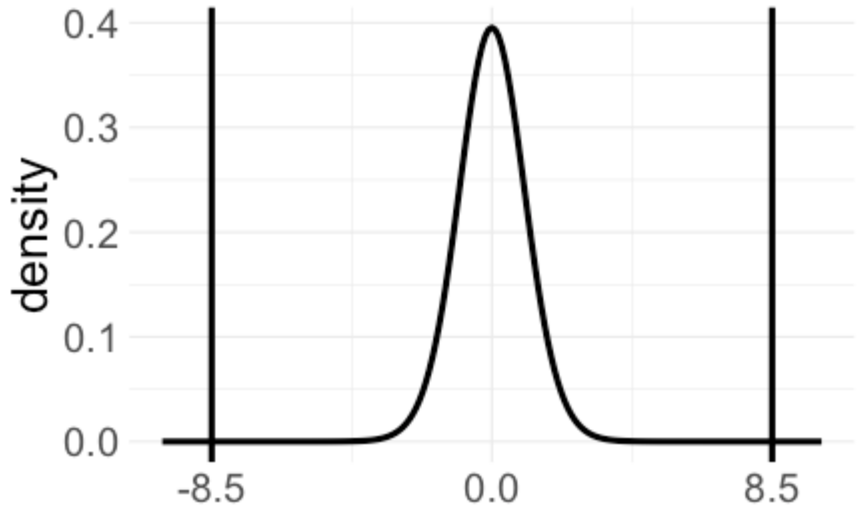
What fraction of the time do we see $t < -8.5$ or $t > 8.5$?

look in left tail

```
pt(-8.5, df=28, lower.tail = TRUE) +
  pt(8.5, df=28, lower.tail = FALSE)
```

look in right tail

How unusual is the observed slope?

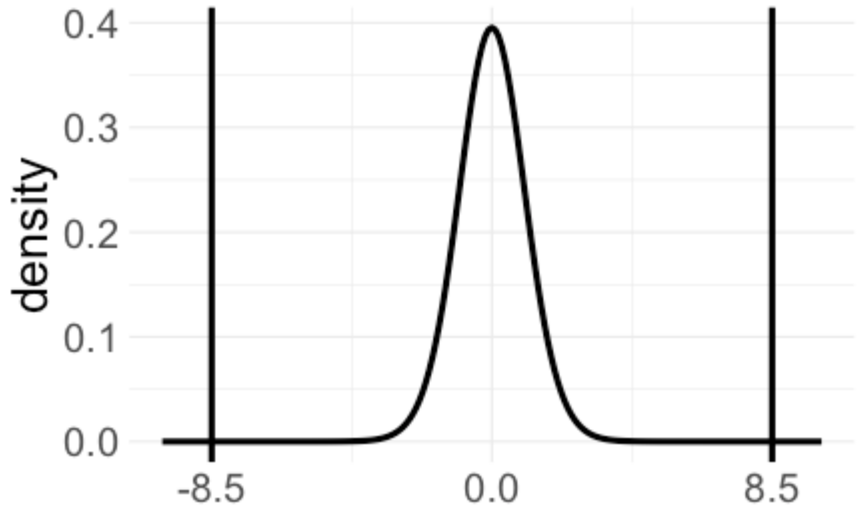



```
pt(-8.5, df=28, lower.tail = TRUE) +
  pt(8.5, df=28, lower.tail = FALSE)
```

```
## [1] 3.056239e-09
```

If $\beta_1 = 0$, the probability of observing $t < -8.5$ or $t > 8.5$ is approximately 0.0000000003

How unusual is the observed slope?



- *If* $\beta_1 = 0$, the probability of observing $t < -8.5$ or $t > 8.5$ is approximately 0.0000000003
- *p-value* = 0.000000003

"p-value = probability of the observed data / observed test statistic, or more extreme, if H_0 were true"

Putting it together

Is there a relationship between mileage and price for used Honda Accords?

Model: $\text{price} = \beta_0 + \beta_1 \text{mileage} + \varepsilon$

Question: Is $\beta_1 = 0$?

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Putting it together

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Estimated line: $\widehat{\text{price}} = 20.8 - 0.12 \text{ mileage}$

Test statistic: $t = \frac{\widehat{\beta}_1}{SE_{\widehat{\beta}_1}} = \frac{-0.12}{0.014} = -8.5$

p-value: 0.000000003

Conclusion: The observed slope of -0.12 would be <u>very</u> unusual if there were no relationship between mileage and price for used Honda Accords, so we have strong evidence that there is a relationship between mileage and price.

In R

```r
accord_lm <- lm(Price ~ Mileage, data = AccordPrice)
summary(accord_lm)
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096     0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198     0.0141   -8.50 3.06e-09 ***
```

- R automatically reports p-values for tests of the form

$$H_0 : \beta_1 = 0 \qquad H_A : \beta_1 \neq 0$$

- *Don't* rely on R output for other tests, such as
  - $H_0 : \beta_1 = 1 \qquad H_A : \beta_1 \neq 1$
  - $H_0 : \beta_1 = 0 \qquad H_A : \beta_1 > 0$

Class activity

Work on the activity (handout), then we will discuss as a class.

Class activity

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

Research question: is there a relationship between sparrow weight and wing length?

> What are the null and alternative hypotheses?

$H_0: \beta_1 = 0$ (no hats in our hypothesis)

$H_A: \beta_1 \neq 0$

Class activity

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$

```
Coefficients:
            Estimate Std. Error
(Intercept)  8.75465    1.39601
Weight       1.31341    0.09756
```

> What is my test statistic for these hypotheses?

$\hat{\beta}_1$

value under H_0

$$t = \frac{1.31341 - 0}{0.09756}$$

$SE_{\hat{\beta}_1}$

$= 13.46$

Class activity

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$

```
Coefficients:
            Estimate Std. Error
(Intercept)  8.75465    1.39601
Weight       1.31341    0.09756
```

$$t = \frac{\widehat{\beta}_1 - 0}{SE_{\widehat{\beta}_1}} = 13.46$$

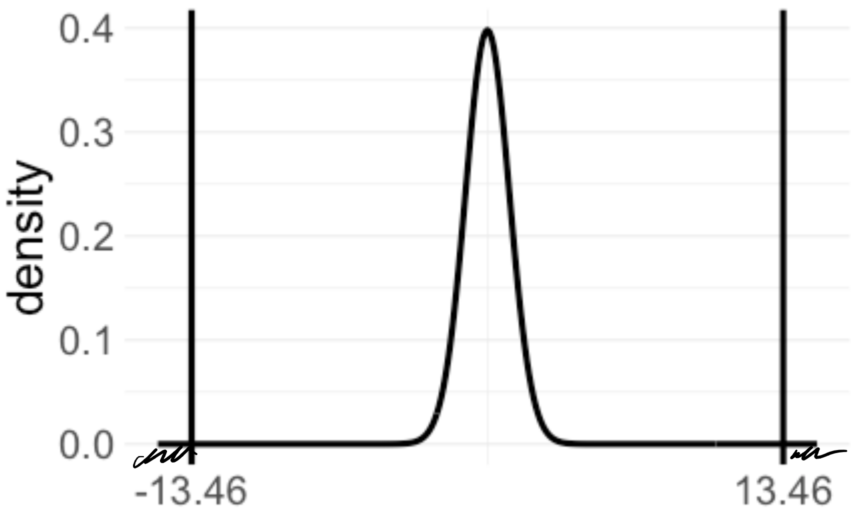
our estimated slope (1.313) is 13.46 standard deviations away from value under H_0 (0)

> Is this an "unusual" value of t if H_0 is true?

very unusual!

Class activity

$t = 13.46 \quad n - 2 = 114$




```
pt(-13.46, 114, lower.tail = TRUE) +
  pt(13.46, 114, lower.tail=FALSE)
```

```
## [1] 2.663963e-25
```

$p \approx 0$